Lecture 25

The Newton kernel and outgoing waves

Objectives. *An exponentially damped radial kernel gives fundamental solutions in three dimensions. Its Fourier transform proves the weak equation. Removing the damping gives the Newton kernel or an outgoing Helmholtz wave, with the side of the distributional boundary value fixed by the damping sign.*

By the end of this lecture you will be able to:

- Compute the Fourier transform of the damped radial kernel.
- Prove its fundamental-solution identity using weak derivatives.
- Derive the outgoing multiplier and its delta contribution on a sphere.
- Solve a compactly supported smooth forcing equation by convolution.

25.1 Weak derivatives in three dimensions

Recall from Definition 7.2 that $\mathcal{S}(\mathbb{R}^3)$ consists of smooth functions whose derivatives decrease faster than every inverse power. Its seminorms can be written $\sup_x (1+|x|)^N |\partial^\alpha \varphi(x)|$, for integers $N \geq 0$ and multi-indices $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ of nonnegative integers. Here $|x|$ is the Euclidean norm and ∂^α means the corresponding partial derivative. The space $\mathcal{S}'(\mathbb{R}^3)$ is its continuous linear dual, with bilinear pairing $T[\varphi]$, as on the line in Lecture 8. Continuity means a bound by a constant times a finite sum of these seminorms, as in Proposition 8.1. The finite-neighbourhood scaling proof there applies to these indices without change. No complex conjugation enters this pairing.

These seminorms define the same topology as the polynomial-weight seminorms in Definition 7.2. Indeed, $|x^\beta| \leq (1+|x|)^{|\beta|}$ bounds each polynomial-weight seminorm by one of the present family. Conversely, $|x| \leq |x_1| + |x_2| + |x_3|$. Expanding the integer power $(1+|x_1|+|x_2|+|x_3|)^N$ bounds each present seminorm by a finite sum of polynomial-weight seminorms. Thus the continuity estimates from Lecture 7 apply to this family. For a polynomial p, multiplication is $(pT)[\varphi] = T[p\varphi]$. It is continuous because the product rule bounds every Schwartz seminorm of $p\varphi$ by finitely many seminorms of φ.

The *weak derivative* is defined by $(\partial_j T)[\varphi] = -T[\partial_j \varphi]$. It is continuous because differentiating a test function shifts its seminorm indices. The Laplacian is $\Delta = \sum_{j=1}^3 \partial_j^2$. The Dirac distribution at the origin is $\delta[\varphi] = \varphi(0)$. A *fundamental solution* of an operator L is a distribution E satisfying $LE = \delta$.

Use the unitary spatial transform from Lecture 7:

$$\widehat{f}(k) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-\mathrm{i}k\cdot x} f(x)\, \mathrm{d}^3x, \qquad \widehat{T}[\varphi] = T[\widehat{\varphi}].$$

The Schwartz transform is a continuous bijection by Theorems 7.3 and 7.5. Thus this definition is continuous and injective on distributions. Differentiation under its test integral gives $\widehat{\partial_j T} = \mathrm{i}k_j\widehat{T}$ and $\widehat{-\Delta T} = |k|^2\widehat{T}$. Indeed $\widehat{\mathrm{i}k_j\varphi} = -\partial_j\widehat{\varphi}$, which is exactly the sign in the weak derivative definition. Also $\widehat{\delta}$ is the constant distribution $(2\pi)^{-3/2}$. These are integration-by-parts identities, not pointwise derivatives of a singular kernel.

25.2 The damped radial kernel

For spherical coordinates let θ_1 and θ_2 be geometric angles with $0 \le \theta_1 \le \pi$ and $0 \le \theta_2 \le 2\pi$. Define the unit direction by

$$e(\theta_1, \theta_2) = \begin{pmatrix} \sin\theta_1 \cos\theta_2 \\ \sin\theta_1 \sin\theta_2 \\ \cos\theta_1 \end{pmatrix}.$$

Define surface measure $\mathrm{d}S$ on S^2 by

$$\int_{S^2} f(e)\,\mathrm{d}S(e) = \int_0^{2\pi}\int_0^{\pi} f(e(\theta_1,\theta_2))\sin\theta_1\,\mathrm{d}\theta_1\,\mathrm{d}\theta_2$$

for nonnegative measurable f, and for absolutely integrable complex f. The spherical map $x = re(\theta_1, \theta_2)$ has absolute Jacobian $r^2\sin\theta_1$. The general change-of-variables formula M5, Theorem 4.2, therefore gives the volume element $\mathrm{d}^3x = r^2\,\mathrm{d}r\,\mathrm{d}S(e)$. The omitted polar axis and angular seam have three-dimensional measure zero. In particular, the surface area is $\int_{S^2}\mathrm{d}S = 2\pi\int_0^{\pi}\sin\theta_1\,\mathrm{d}\theta_1 = 4\pi$.

The decay parameter η has the same role as in Lecture 8. For $\operatorname{Re}\eta > 0$ define the radial function

$$E_\eta(x) = \frac{e^{-\eta r}}{4\pi r}, \qquad r = |x| > 0. \tag{25.1}$$

Its value at the origin does not matter for integration. Near zero, its radial absolute integral is bounded by a constant times $\int_0^1 r\,\mathrm{d}r$. At infinity the exponential is integrable. Thus $E_\eta \in L^1(\mathbb{R}^3)$ and defines a tempered distribution.

Theorem 25.1. *For* $\operatorname{Re}\eta > 0$,

$$\widehat{E_\eta}(k) = \frac{(2\pi)^{-3/2}}{|k|^2 + \eta^2}, \qquad (-\Delta + \eta^2)E_\eta = \delta. \tag{25.2}$$

Proof. Rotation preserves the radial factors and the volume and spherical measures: by M5 its Jacobian has absolute determinant one. For $k \neq 0$, take the polar axis in the direction of k and put $K = |k| > 0$. Fubini and spherical change of variables are the standing facts M4 and M5, Theorems 4.1 and 4.2. Absolute integrability was just proved. Let $e \in S^2$ denote a unit direction, with surface measure $\mathrm{d}S(e)$. Writing u for the cosine of its polar angle gives the angular integral

$$\int_{S^2} e^{-\mathrm{i}Kru}\,\mathrm{d}S(e) = 2\pi\int_{-1}^{1} e^{-\mathrm{i}Kru}\,\mathrm{d}u = 4\pi\frac{\sin(Kr)}{Kr}.$$

The unit sphere has area 4π. The radial integral becomes

$$\widehat{E}_\eta(k) = (2\pi)^{-3/2} \int_0^\infty e^{-\eta r} \frac{\sin(Kr)}{K}\, \mathrm{d}r = \frac{(2\pi)^{-3/2}}{\eta^2 + K^2}.$$

To verify the last step, write the sine as two exponentials. Their integrals are $1/(\eta - \mathrm{i}K)$ and $1/(\eta + \mathrm{i}K)$. Their difference divided by $2\mathrm{i}K$ is $1/(\eta^2 + K^2)$. Both exponentials decay since $\operatorname{Re}\eta > 0$. At $k = 0$ the integral is $(2\pi)^{-3/2} \int_0^\infty re^{-\eta r}\, \mathrm{d}r = (2\pi)^{-3/2}\eta^{-2}$, the same formula. The denominator cannot vanish for real k when $\operatorname{Re}\eta > 0$.

For an integrable function, Fubini shows that the distributional Fourier transform equals its ordinary transform. Multiplying that transform by $|k|^2 + \eta^2$ gives $(2\pi)^{-3/2} = \widehat{\delta}$. The weak derivative rule and injectivity of the distributional transform therefore prove the claimed equation, including its contribution at zero. ∎

For $r > 0$ direct differentiation gives $-E_\eta'' - 2E_\eta'/r + \eta^2 E_\eta = 0$. This pointwise identity alone would miss the Dirac distribution at the origin. The weak argument determines its coefficient.

25.3 The Newton limit

Let $\eta > 0$ be real and decrease it to zero. For every Schwartz test function, $|E_\eta(x)\varphi(x)| \le |\varphi(x)|/(4\pi|x|)$. This majorant is integrable: $1/|x|$ is locally integrable near the origin in three dimensions, and the test function controls infinity. Dominated convergence therefore gives convergence in $\mathcal{S}'$ to

$$E_0(x) = \frac{1}{4\pi|x|}, \qquad -\Delta E_0 = \delta. \tag{25.3}$$

The weak equation follows by testing the damped equation and taking limits. The term $\eta^2 E_\eta[\varphi]$ tends to zero by the same bound. In Fourier variables the reciprocals of the denominators are bounded by $1/|k|^2$. This is locally integrable in dimension three and integrable against Schwartz decay at infinity. Thus $\widehat{E}_0 = (2\pi)^{-3/2}/|k|^2$ as a regular distribution.

The Newton potential of a compactly supported smooth density f is $u = E_0 * f$. The convolution argument below proves $-\Delta u = f$. It fixes one solution. Adding a harmonic function on all of $\mathbb{R}^3$ would give another solution without a decay or boundary requirement.

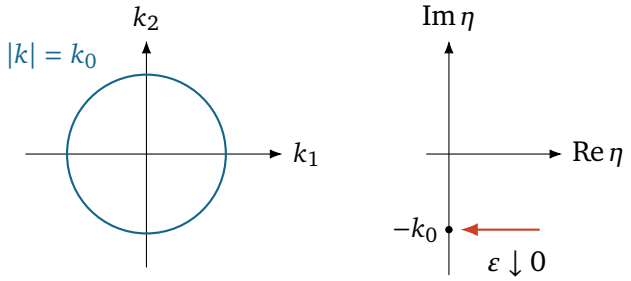
25.4 The outgoing boundary value

Fix a wave number $k_0 > 0$ and set $\eta = \varepsilon - \mathrm{i}k_0$ with $\varepsilon > 0$. This gives

$$E_\eta(x) = \frac{e^{-\varepsilon r} e^{\mathrm{i}k_0 r}}{4\pi r}, \qquad |k|^2 + \eta^2 = |k|^2 - k_0^2 + \varepsilon^2 - 2\mathrm{i}\varepsilon k_0.$$

The same test-function majorant proves convergence in $\mathcal{S}'$ to the *outgoing kernel* $E_{\mathrm{out}}(x) = e^{\mathrm{i}k_0 r}/(4\pi r)$. The weak equations pass to the limit and give $(-\Delta - k_0^2)E_{\mathrm{out}} = \delta$. The choice of damping fixes a side of approach to the real denominator.

The singular frequencies lie on a sphere, rather than at the origin.



A planar section of the frequency sphere is shown beside the approach $\eta = \varepsilon - \mathrm{i}k_0$ from the right half-plane.

Define the shell distribution by

$$\delta(|k|^2 - k_0^2)[\varphi] = \frac{1}{2k_0}\int_{|k|=k_0} \varphi(k)\,\mathrm{d}S(k).$$

Surface measure on the radius-k_0 sphere is defined by rescaling:

$$\begin{aligned}&\int_{|k|=k_0} \varphi(k)\,\mathrm{d}S(k)\\ &= k_0^2 \int_{S^2} \varphi(k_0 e)\,\mathrm{d}S(e).\end{aligned}$$

Define the principal value by symmetric exclusion of $\big||k|^2 - k_0^2\big| < h$ and the limit $h \downarrow 0$.

Proposition 25.2. *The transform of the outgoing kernel is*

$$\begin{aligned}\widehat{E}_{\mathrm{out}} = (2\pi)^{-3/2}\,\mathrm{PV}\,\frac{1}{|k|^2 - k_0^2}\\ + (2\pi)^{-3/2}\mathrm{i}\pi\delta(|k|^2 - k_0^2).\end{aligned} \tag{25.4}$$

The sum after factoring out $(2\pi)^{-3/2}$ is denoted $(|k|^2 - k_0^2 - \mathrm{i}0)^{-1}$.

Proof. Take a smooth radial cutoff supported near the sphere and equal to one on a smaller neighbourhood. Away from this support the denominators stay away from zero for small ε, and ordinary dominated convergence applies against Schwartz tests. On the shell put $u = r^2 - k_0^2$. Spherical coordinates turn the test integral into a one-dimensional integral with compactly supported smooth numerator

$$g(u) = \frac{\sqrt{u + k_0^2}}{2}\int_{S^2} \varphi\big(\sqrt{u + k_0^2}\,e\big)\,\mathrm{d}S(e)$$

multiplied by that cutoff. The numerator used in the following estimate and boundary-value formula is the product just formed. Its support is away from $u = -k_0^2$. First remove the real shift ε^2 in the denominator. With $d = 2\varepsilon k_0$ the difference is bounded in absolute integral by

$$\begin{aligned}&\|g\|_\infty\,\varepsilon^2 \int_{\mathbb{R}} \frac{\mathrm{d}u}{\sqrt{(u+\varepsilon^2)^2 + d^2}\sqrt{u^2 + d^2}}\\ &\le \|g\|_\infty\,\frac{\pi\varepsilon^2}{d} \longrightarrow 0.\end{aligned}$$

The inequality is Cauchy–Schwarz and $\int_{\mathbb{R}} (u^2 + d^2)^{-1}\,\mathrm{d}u = \pi/d$. The Plemelj lemma, Theorem 8.2, now gives PV $\int g(u)/u\,\mathrm{d}u + \mathrm{i}\pi g(0)$. The value $g(0)$ is precisely the action of the shell distribution defined above on the test function. Combining shell and complement proves the formula and existence of the stated principal value. Continuity of the Fourier transform transfers the already proved spatial limit to this multiplier. ∎

This kernel satisfies the radial outgoing condition

$$r(\partial_r - \mathrm{i}k_0)E_{\mathrm{out}} = -\frac{e^{\mathrm{i}k_0 r}}{4\pi r} \longrightarrow 0.$$

It is a direct property of this kernel. No general radiation-condition uniqueness theorem is needed here. Choosing $\eta = \varepsilon + \mathrm{i}k_0$ instead gives the incoming phase $e^{-\mathrm{i}k_0 r}$ and the multiplier with +i0, whose shell term is $-\mathrm{i}\pi\delta$.

25.5 Compact smooth forcing

Let E be any of the kernels above and $f \in C_c^\infty(\mathbb{R}^3)$. Define $u(x) = \int E(y) f(x-y)\,\mathrm{d}^3 y$. The kernel is locally integrable. On compact sets of x, the integration uses a fixed bounded set of y, so differentiation may be passed to f by dominated convergence. Thus u is smooth.

For a compactly supported smooth test function φ, Fubini gives

$$\int u(x)(L\varphi)(x)\,\mathrm{d}^3 x = \int f(t)\left(\int E(y)(L\varphi)(y+t)\,\mathrm{d}^3 y\right)\mathrm{d}^3 t = \int f(t)\varphi(t)\,\mathrm{d}^3 t.$$

Here $L = -\Delta + \eta^2$ or its limiting operator. The inner identity is the weak fundamental-solution equation applied to $y \mapsto \varphi(y+t)$. Absolute Fubini holds on a bounded integration region using local integrability of E. Hence $Lu = f$ weakly and, since u is smooth, pointwise.

25.6 Comparison with evolution kernels

An *initial-data kernel* propagates data prescribed at time zero. It solves a different problem from a stationary spatial fundamental solution. The following comparison is cited from Strichartz, Ch. 5. The equations in this comparison have dimensionless coefficients.

- The heat equation $\partial_t u = \Delta u$ with $u(0) = f$ has kernel
$$(4\pi t)^{-3/2} e^{-r^2/(4t)}.$$
- The wave equation $\partial_t^2 u = \Delta u$ with $u(0) = 0$ and $u_t(0) = f$ has kernel
$$\frac{\delta(r-t)}{4\pi r}.$$
- The Schrödinger equation $\mathrm{i}\partial_t u = -\Delta u$ with $u(0) = f$ has kernel
$$(4\pi \mathrm{i} t)^{-3/2} e^{\mathrm{i}r^2/(4t)}.$$

The wave kernel is a radial distribution with spatial volume measure. The Schrödinger power uses $\mathrm{i}^{3/2} = e^{3\pi\mathrm{i}/4}$ for $t > 0$. Multiplication of the wave kernel by the positive-time indicator produces the retarded spacetime fundamental solution of $\partial_t^2 - \Delta$. For the other two equations, a spacetime fundamental solution requires the normalisation of the chosen time operator. The comparison asserts initial data only.

Summary.

- The damped radial transform proves $(-\Delta + \eta^2)E_\eta = \delta$ for $\operatorname{Re}\eta > 0$, including the origin.
- Real $\eta \downarrow 0$ gives the Newton kernel $1/(4\pi r)$.
- The outgoing limit $\eta = \varepsilon - \mathrm{i}k_0$ gives phase $e^{\mathrm{i}k_0 r}$ and denominator $|k|^2 - k_0^2 - \mathrm{i}0$. Its shell term has positive imaginary sign.
- Convolution with a compact smooth forcing gives a smooth weak solution.

The side of a boundary value also controls the half-plane analyticity used in Lecture 26 and Lecture 27.

References

R. Strichartz, *A Guide to Distribution Theory and Fourier Transforms*, CRC, 1994, §§5.1–5.4, pp. 56–69. His forward transform has positive exponent and no unitary prefactor; our spatial transform has negative exponent and factor $(2\pi)^{-3/2}$. His Newton sign for Δ is reversed here for $-\Delta$. G. Teschl, *Mathematical Methods in Quantum Mechanics*, online version 12 February 2009, §7.4, pp. 171–172, formula (7.48), gives the same resolvent kernel with $\eta = \sqrt{-z}$ on the decaying branch.